

SUMMATIVE ASSESSMENT: EVALUATING LEARNING OUTCOMES IN THE FORM OF TERMINAL/FINAL EXAMINATION

Summative assessment is a fundamental component of the educational process, designed to evaluate student learning at the conclusion of an instructional period. Unlike formative assessment, which provides ongoing feedback during the learning process, summative assessment serves as a final measure of what students have learned. Typically administered at the end of a unit, course, or academic year. Summative assessment aims to determine the extent to which educational objectives have been achieved.

The primary purpose of summative assessment is to assess the overall effectiveness of instruction and learning. It provides a conclusive evaluation of student performance, often in the form of tests, final projects, or standardized exams. These assessments generate grades or scores that reflect a student's achievement in a given subject area over a specific period or time duration.

Summative assessment is often used to make critical decisions regarding student progression, certification, or placement in subsequent educational levels. Additionally, summative assessments provide valuable data that inform curriculum development and instructional strategies. By analyzing summative assessment results, educators can identify trends, strengths, and weaknesses within their instructional approaches, allowing for improvements in future teaching.

In conclusion, summative assessment plays a critical role in the educational process by providing a final evaluation of student learning. While it differs from formative assessment in its focus and application, it is an essential tool for measuring academic achievement. When balanced with formative assessments, summative assessments contribute to a well-rounded and effective approach to evaluating and supporting student learning.

National Curriculum of Pakistan 2022-2023
ASSESSMENT FRAME WORK BIOLOGY Grade-IX (SSC I)
Details of Content Areas/ SLOs

Content Domain / Area	SLO No./ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods Required (1 period=40 minutes)
A	[SLO: B-09-A-01] Define biology	Summative	Knowledge	Question(s) will be asked in annual examination.	10 periods
	[SLO: B-09-A-02] State Quran instructs to reveal the study of Life	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-A-03] Define major fields of biology as Botany, zoology and Microbiology	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-A-04] Define with examples that biology has many sub-fields. (Cytology) (Embryology) (Genetics) (Molecular Biology) (Pathology) (Ecology) (Marine Biology) (Immunology) (Morphology) (Anatomy) (Histology) (Physiology) (Taxonomy) (Palaeontology) (Pharmacology)	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-A-05] Relate that biology connects with other natural sciences. Students should be able to distinguish in terms of the broad subject matter the below fields: (Biophysics) (Biochemistry) (Computational Biology) (Biogeography) (Biostatistics) (Biotechnology) (Bio economics)	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-A-06] Identify the careers in Biology and Explain with examples how biology is a subset of the natural sciences and of the life sciences.	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-A-07] Justify with examples that science is a collaborative field that requires interdisciplinary researchers working together to share knowledge and critique ideas	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-A-08] Describe the steps of the scientific method that is: Recognition Observation Hypothesis Deduction Experiments Results	Summative	Understanding	Question(s) will be asked in annual examination.	

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	SLO: B-09-A-09] Evaluate the terms 'hypothesis', 'theory' and 'law' in the context of research in the natural sciences	Summative	Application	Question(s) will be asked in annual examination.	
B	[SLO: B-09-B-01] Explain the theory of evolution by natural selection with example	Summative	Understanding	Question(s) will be asked in annual examination.	20 periods
	[SLO: B-09-B-02] Define Species	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-03] Describe speciation	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-04] Discuss briefly the observations Darwin made during his voyage on HMS Beagle	Formative	Understanding	This SLO is part of regular teaching and learning but will not be assessed in annual examination. (As numerous observations were done in this voyage so it is included in formative assessment.)	
	[SLO: B-09-B-05] Describe sources of variation which can lead to speciation and evolution	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-06] Describe evidence of evolution with regards to the following - Palaeontology (fossil record) - Comparative anatomy (homologous structures, vestigial structures) - Selective breeding	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-07] Define biodiversity and classification	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-08] Describe advantages of classification	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-09] Discuss the history of classification schemes	Summative	Understanding	Question(s) will be asked in annual examination.	

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	[SLO: B-09-B-10] List the three distinct domains into which living organisms are broadly classified into	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-11] List the taxonomic ranks of classification	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-12] Outline the binomial nomenclature system	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-B-13] Describe the complications of classifying viruses	Summative	Understanding	Question(s) will be asked in annual examination.	
C	[SLO: B-09-C-1] Define Biochemistry/molecular biology	Summative	Knowledge	Question(s) will be asked in annual examination.	20 periods
	[SLO: B-09-C-2] Outline the various types of common biomolecules (DNA, RNA, Proteins, Lipids, Carbohydrates) including their locations inside the cell and main roles	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-C-3] Outline the structure and function and sources of proteins with structure of amino acids	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-C-4] Outline the structure, function and sources of lipids	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-C-05] Define Carbohydrates and Outline the structure, function and sources of Carbohydrates	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-C-06] Identify carbohydrates as monosaccharides, disaccharides and polysaccharides	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-C-07] Describe briefly the structure of DNA as a double helix macromolecule made of nucleotides with base pairing in between the two helices through complementary base pairing	Summative	Application	Question(s) will be asked in annual examination.	

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	[SLO: B-09-C-08] Outline function of DNA as carrier of, hereditary information	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-C-09] Describe briefly the structure of RNA as single stranded macromolecule made of nucleotides with nitrogenous base overhangs	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-C-10] Outline the function of RNA as aid in converting hereditary information into useful proteins	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-C-11] Outline how information in the DNA is converted to information on RNA and then into proteins	Summative	Understanding	Question(s) will be asked in annual examination.	
D	[SLO: B-09-D-1] Describe cell as the basic unit of life	Summative	Understanding	Question(s) will be asked in annual examination.	28 periods
	[SLO: B-09-D-2] Compare with diagrams the structure of animal and plant cells	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-D-3] Sketch different sub-cellular organelles (nucleus, mitochondria, cell membranes, etc.) and outline their roles	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-D-4] Outline structural advantages of plant and animal cells	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-D-05] Identify different types of cells (mesophyll cell, epidermal cell, neurons, muscle, red blood cell, liver cell) and sketch their structures	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-D-06] Describe the concept of division of labour and how it applies to - within cells (across subcellular organelles) - multicellular organisms (across cells)	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-D-07] Describe Cell Specialization.	Summative	Understanding	Question(s) will be asked in annual examination.	

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	[SLO: B-09-D-08] Describe Cell cycle	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-D-09] Explain mitosis, meiosis and stages of mitosis, meiosis (by use of sketch and diagrams)	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-D-11] Compare the processes of mitosis and meiosis	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-D-12] Outline the significance of mitosis and meiosis	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-D-13] Define stem cells as unspecialized cell	Summative	Understanding	Question(s) will be asked in annual examination.	
E	[SLO: B-09-E-1] Distinguish between tissues, organs and system with examples from animals and plants	Summative	Understanding	Question(s) will be asked in annual examination.	8 periods
	[SLO: B-09-E-2] Describe the concept of emergent properties as gain in functionalities and how it applies to the following going from sub-cellular organelles to cells - going from cells to tissues - going from tissues to organs - going from organs to systems - going from systems to living organisms	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-E-3] Enlist the different types of tissue come together to form the stomach organ in the human body	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-E-4] Discuss the different types of tissue come together to form the leaf	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-E-5] Discuss the organ system come together to form the human body	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-E-06] Describe the advantages of homeostasis	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-E-07] Discuss the various organs and systems of the human body work to maintain homeostasis	Summative	Understanding	Question(s) will be asked in annual examination.	

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	[SLO: B-09-E-08] Explain plant physiology in terms of structures and roles of various plant organs	Summative	Understanding	Question(s) will be asked in annual examination.	
F	[SLO: B-09-F-01] Define metabolism, catabolism and anabolism with examples	Summative	Understanding	Question(s) will be asked in annual examination.	18 periods
	[SLO: B-09-F-02] Define Enzymes and describe their characteristics	Summative	Knowledge/ understanding	Question(s) will be asked in annual examination. This SLO is compound SLO. Both levels can be asked.	
	[SLO: B-09-F-03] Show the mechanism of enzyme action	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-F-04] Assess the factors which could influence enzyme activity	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-F-05] Describe competitive, and non-competitive inhibition	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-F-06] Discuss the role of ATP as energy currency	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-F-07] Describe photosynthesis in plants	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-F-08] Explain aerobic respiration and anaerobic respiration	Summative	Understanding	Question(s) will be asked in annual examination.	
Q	[SLO: B-09-Q-01] Define mineral nutrition in plants	Summative	Knowledge	Question(s) will be asked in annual examination.	40 periods
	[SLO: B-09-Q-02] Categorize minerals nutrients of plants into macronutrients and micronutrients	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-03] State that nitrogen is important in protein synthesis and magnesium for chlorophyll formation	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-04] Conceptualize transport and its needs	Summative	Application	Question(s) will be asked in annual examination.	

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	[SLO: B-09-Q-05] Explain the internal structure of root and root hair	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-06] Describe how roots take up water and mineral salts by active and passive absorption	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-07] Describe transpiration and relate this process with cell surface and stomatal opening and closing	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-08] Describe temperature, wind and humidity as the factors affecting the rate of transpiration	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	
	[SLO: B-09-Q-09] Describe the mechanism of transport of water and salt in plants	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-10] Explain the mechanism of food translocation by, the theory of Pressure Flow Mechanism	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-11] Describe the process of gaseous exchange in plants	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-12] Define homeostasis and describe its importance	Summative	Knowledge/ understanding	Question(s) will be asked in annual examination. This SLO is compound SLO. Both levels can be asked.	
	[SLO: B-09-Q-13] Describe the mechanism and adaptations in plants for the excretion	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-14] Explain osmotic adjustments in plants	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-15] Describe different types of asexual reproduction i.e. binary fission, budding, spore formation and vegetative propagation	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-16] Distinguish between vegetative propagation and artificial propagation	Summative	Application	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-17] Explain vegetative propagation in plants (through stem, suckers and leaves)	Summative	Understanding	Question(s) will be asked in annual examination.	

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	[SLO: B-09-Q-18] Describe the two methods of artificial vegetative propagation (stem cuttings and grafting)	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-19] Rationalize how parthenogenesis is a type of asexual reproduction	Summative	Understanding	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-20] Define cloning	Summative	Knowledge	Question(s) will be asked in annual examination.	
	[SLO: B-09-Q-21] Explain sexual reproduction in plants	Summative	Knowledge/ understanding/ Application	Question(s) will be asked in annual examination. This SLO is compound SLO. All three levels can be asked.	

PRACTICAL SLOs

Domain	NCP SLOs Description	Form of Assessment	Cognitive Level	Remarks	Number of Periods Required (1 period=40 minutes)
X	[SLO: B-09-10-X-01] Students should be able to simple measurements in SI Units of: - volumes of gases or solutions/liquids - masses - temperatures - times - lengths	Summative for PBA	Application	Laboratory work- will be assessed in PBA.	20 periods
	[SLO: B-09-10-X-02] Students should be able to carry out simple experiments of: - diffusion - osmosis - food tests	Summative for PBA	Application	Laboratory work- will be assessed in PBA.	